

## 8.7 Network-Layer Security: IPsec and Virtual Private Networks

The IP security protocol, more commonly known as **IPsec**, provides security at the network layer. IPsec secures IP datagrams between any two network-layer entities, including hosts and routers. As we will soon describe, many institutions (corporations, government branches, non-profit organizations, and so on) use IPsec to create **virtual private networks (VPNs)** that run over the public Internet.

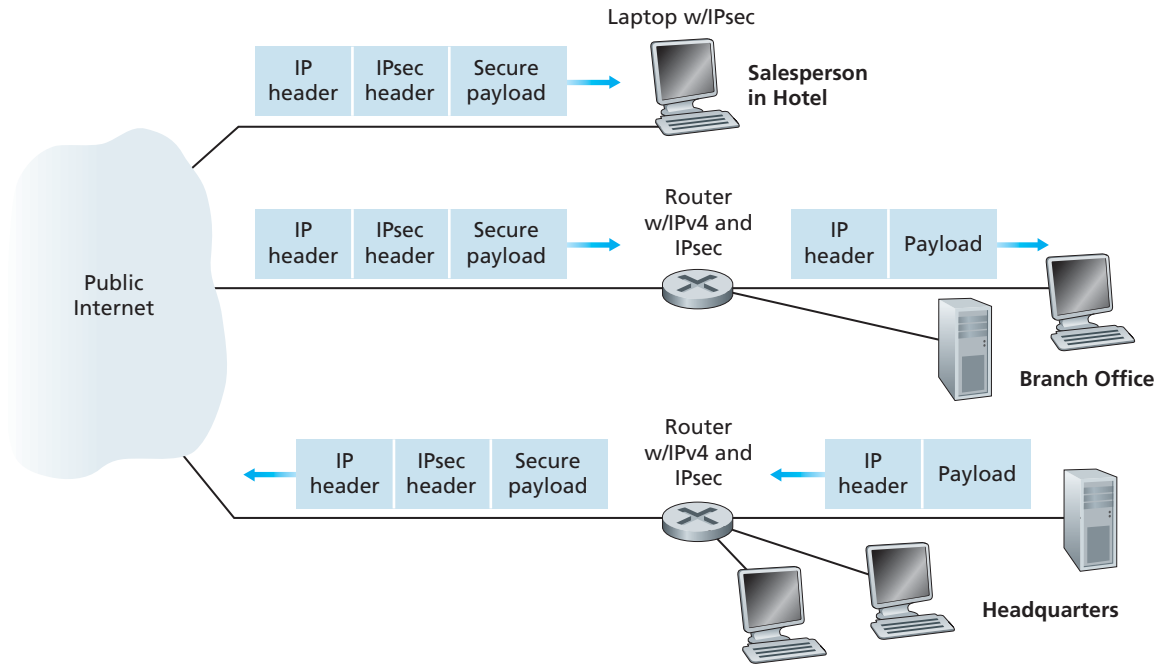
Before getting into the specifics of IPsec, let's step back and consider what it means to provide confidentiality at the network layer. With network-layer confidentiality between a pair of network entities (for example, between two routers, between two hosts, or between a router and a host), the sending entity encrypts the payloads of all the datagrams it sends to the receiving entity. The encrypted payload could be a TCP segment, a UDP segment, an ICMP message, and so on. If such a network-layer service were in place, all data sent from one entity to the other—including e-mail, Web pages, TCP handshake messages, and management messages (such as ICMP and SNMP)—would be hidden from any third party that might be sniffing the network. For this reason, network-layer security is said to provide “blanket coverage”.

In addition to confidentiality, a network-layer security protocol could potentially provide other security services. For example, it could provide source authentication, so that the receiving entity can verify the source of the secured datagram. A network-layer security protocol could provide data integrity, so that the receiving entity can check for any tampering of the datagram that may have occurred while the datagram was in transit. A network-layer security service could also provide replay-attack prevention, meaning that Bob could detect any duplicate datagrams that an attacker might insert. We will soon see that IPsec indeed provides mechanisms for all these security services, that is, for confidentiality, source authentication, data integrity, and replay-attack prevention.

### 8.7.1 IPsec and Virtual Private Networks (VPNs)

An institution that extends over multiple geographical regions often desires its own IP network, so that its hosts and servers can send data to each other in a secure and confidential manner. To achieve this goal, the institution could actually deploy a stand-alone physical network—including routers, links, and a DNS infrastructure—that is completely separate from the public Internet. Such a disjoint network, dedicated to a particular institution, is called a **private network**. Not surprisingly, a private network can be very costly, as the institution needs to purchase, install, and maintain its own physical network infrastructure.

Instead of deploying and maintaining a private network, many institutions today create VPNs over the existing public Internet. With a VPN, the institution's inter-office traffic is sent over the public Internet rather than over a physically independent network. But to provide confidentiality, the inter-office traffic is encrypted before it enters the public Internet. A simple example of a VPN is shown in Figure 8.27. Here



**Figure 8.27** ♦ Virtual Private Network (VPN)

the institution consists of a headquarters, a branch office, and traveling salespersons that typically access the Internet from their hotel rooms. (There is only one salesperson shown in the figure.) In this VPN, whenever two hosts within headquarters send IP datagrams to each other or whenever two hosts within the branch office want to communicate, they use good-old vanilla IPv4 (that is, without IPsec services). However, when two of the institution's hosts communicate over a path that traverses the public Internet, the traffic is encrypted before it enters the Internet.

To get a feel for how a VPN works, let's walk through a simple example in the context of Figure 8.27. When a host in headquarters sends an IP datagram to a salesperson in a hotel, the gateway router in headquarters converts the vanilla IPv4 datagram into an IPsec datagram and then forwards this IPsec datagram into the Internet. This IPsec datagram actually has a traditional IPv4 header, so that the routers in the public Internet process the datagram as if it were an ordinary IPv4 datagram—to them, the datagram is a perfectly ordinary datagram. But, as shown Figure 8.27, the payload of the IPsec datagram includes an IPsec header, which is used for IPsec processing; furthermore, the payload of the IPsec datagram is encrypted. When the IPsec datagram arrives at the salesperson's laptop, the OS in the laptop decrypts the payload (and provides other security services, such as verifying data integrity) and passes the unencrypted payload to the upper-layer protocol (for example, to TCP or UDP).

We have just given a high-level overview of how an institution can employ IPsec to create a VPN. To see the forest through the trees, we have brushed aside many important details. Let's now take a closer look.

## 8.7.2 The AH and ESP Protocols

IPsec is a rather complex animal—it is defined in more than a dozen RFCs. Two important RFCs are RFC 4301, which describes the overall IP security architecture, and RFC 6071, which provides an overview of the IPsec protocol suite. Our goal in this textbook, as usual, is not simply to re-hash the dry and arcane RFCs, but instead take a more operational and pedagogic approach to describing the protocols.

In the IPsec protocol suite, there are two principal protocols: the **Authentication Header (AH)** protocol and the **Encapsulation Security Payload (ESP)** protocol. When a source IPsec entity (typically a host or a router) sends secure datagrams to a destination entity (also a host or a router), it does so with either the AH protocol or the ESP protocol. The AH protocol provides source authentication and data integrity but *does not* provide confidentiality. The ESP protocol provides source authentication, data integrity, *and* confidentiality. Because confidentiality is often critical for VPNs and other IPsec applications, the ESP protocol is much more widely used than the AH protocol. In order to de-mystify IPsec and avoid much of its complication, we will henceforth focus exclusively on the ESP protocol. Readers wanting to learn also about the AH protocol are encouraged to explore the RFCs and other online resources.

## 8.7.3 Security Associations

IPsec datagrams are sent between pairs of network entities, such as between two hosts, between two routers, or between a host and router. Before sending IPsec datagrams from source entity to destination entity, the source and destination entities create a network-layer logical connection. This logical connection is called a **security association (SA)**. An SA is a simplex logical connection; that is, it is unidirectional from source to destination. If both entities want to send secure datagrams to each other, then two SAs (that is, two logical connections) need to be established, one in each direction.

For example, consider once again the institutional VPN in Figure 8.27. This institution consists of a headquarters office, a branch office and, say,  $n$  traveling salespersons. For the sake of example, let's suppose that there is bi-directional IPsec traffic between headquarters and the branch office and bi-directional IPsec traffic between headquarters and the salespersons. In this VPN, how many SAs are there? To answer this question, note that there are two SAs between the headquarters gateway router and the branch-office gateway router (one in each direction); for each salesperson's laptop, there are two SAs between the headquarters gateway router and the laptop (again, one in each direction). So, in total, there are  $(2 + 2n)$  SAs. *Keep in mind, however, that not all traffic sent into the Internet by the gateway routers or by the laptops will be IPsec secured.* For example, a host in headquarters may want to access a Web server (such as Amazon or Google) in the public Internet. Thus, the gateway router (and the laptops) will emit into the Internet both vanilla IPv4 datagrams and secured IPsec datagrams.